CHINMAYA INTERNATIONAL RESIDENTIAL SCHOOL, COIMBATORE

SAMPLE QUESTION PAPER

XI (CBSE)
CHEMISTRY

SECTION A

01. Write the IUPAC name of the following compounds.

a) HCHO b) CH_3- CH_2 – NO_2

02. Find the pH of hydrochloric acid when the hydrogen ion concentration is 0.01M.

03. Write the chemical formula of bleaching powder and baking soda.

SECTION B

04. At equilibrium, the concentrations of $N_2 = 3.0 \times 10^{-3}$M, $O_2 = 4.2 \times 10^{-3}$M and NO = 2.8×10^{-3}M in a sealed vessel at 800 K. Calculate K_c for the reaction.

$$N_2\,(g) + O_2\,(g) \rightleftharpoons 2NO\,(g)$$

05. Give reason:

a) Tartaric acid is added with baking powder in the preparation of breads and cakes.

b) Gypsum is added during the manufacture of cement.

06. How will you distinguish between an aldehyde and a ketone? Give an equation.

07. Complete the following reaction:

a) $CH_3CH_2OH \xrightarrow{\text{alk. } KMnO_4}$

b) $CH_2 = CH_2 + H_2O \xrightarrow[\Delta]{H_2SO_4}$

SECTION C

08. Explain the following terms:

a) Vulcanisation

b) Esterification :

c) Saponification

09. Name an alloy of:

a) Copper used in making statues

b) Aluminium used in the construction of aircrafts.

c) Lead used in joining metals for electrical work.

10.Define rate of a reaction. Consider the following reaction:

$2KBr\ (aq) + Cl_2\ (aq) \longrightarrow 2KCl\ (aq) + Br_2\ (aq)$

The concentration of Br_2 after 15 seconds of the start of the reaction is measured to be 2.0×10^{-4} mol L^{-1}. Calculate the average reaction rate.

SECTION D

11.a) Give an equation for the preparation of hydrogen in the laboratory.

b) Draw a labeled diagram for the above preparation.

c) Name the type of the reaction involved in the above preparation.